



Founder · Archi Cam AI

KOA MARIE GERVAIS NELLY

Lead AI Engineer & Data Architect | Founder @ Archi Cam AI

EXECUTIVE SUMMARY

Applied AI Engineer specializing in autonomous agent architectures, I design deterministic, explainable, and sovereign zero-hallucination systems. Member of the Google Developer Program, I bridge civil engineering rigor with advanced data science to solve complex operational challenges. Founder of Archi Cam AI (5D BIM SaaS & predictive estimation), I build auditable, robust, and high-impact AI solutions.

FLAGSHIP AI PROJECTS

Archi Cam AI — *Agentic AI & 5D BIM Construction SaaS*

- Full automation of construction quantity takeoff and cost estimation: eliminating human error and material waste.
- Instant generation of standardized BOQ estimates in <45s (99.2% time savings) and 3D architectural renders.
- Deterministic zero-hallucination structural concrete validation enforcing strict compliance with urban planning codes.

K1-MATHINFO (v3.2.0) — *Sovereign Academic Research Platform*

- GraphRAG over 30 yrs archives (471 theses DMI, 4,494 Neo4j relations), LatentGate (5.2ms) & vCache (0% FP).
- Tri-engine WRRF (P@5=96.4%), S-GRPO, OKF v0.2 SHA-256 No-LLM, 192 tests (100%), and Dev.to deep-dive.

Sovereign.BI Agentic — *Enterprise Business Intelligence & Decisions*

- Executive decision enablement: querying complex corporate data warehouses directly in natural language in under 5 seconds.
- Total sovereign enterprise privacy with air-gapped local execution and mathematical explainability for every KPI.

Dataset Automator & VigieSahel — *Climate Resilience & Production MLOps*

- Dataset Automator: MLOps data factory (TabFM, PAIR WIT, EU AI Act RSASSA-PSS seals) and Dev.to publication.
- VigieSahel: Sahelian resilience system cutting crop sowing failures by 35% and forecasting meningitis epidemics 14 days ahead.

PROFESSIONAL EXPERIENCE

AI Lead & Data Science Consultant — *March 2026 – Present*

Independent Projects & Enterprises | Douala, CM

- Ethical sovereign AI systems: GraphRAG Neo4j pipelines, high-dimensional EDA, multi-source RAG architectures.
- Executive decision reporting, automated KPI dashboards, and SHAP explainability audits for African SMEs.

Aviation Security Officer (AVSEC) — *2018 – Present*

CCAA — Cameroon Civil Aviation Authority | Douala, CM

- Threat assessment, secure access control, and regulatory compliance audits (ICAO Annex 17).
- Operational crisis management: emergency team coordination, anti-intrusion protocols.

EDUCATION & CERTIFICATIONS

M.Sc. in Applied Artificial Intelligence — *Dec. 2025 – 2027 [In Progress]*

University of Ngaoundéré | Cameroon

- ML & Bayesian Statistics, Data Engineering & Neo4j, Computer Vision & Robotics, Ethics & Cybersecurity, Production MLOps.
- Research project: K1-MATHINFO v3.2 sovereign system — multi-source agent certified OKF v0.2 (SHA-256 No-LLM, 192 tests).

B.Sc. in Civil Engineering (Building Option) — *2015 – 2016*

ISTDI / IUC Douala | Cameroon

- Structural calculations (BAEL 91), quantity surveying, construction project management — AI-applied estimation base.

HONORS & APPLIED AI DISTINCTIONS

- CCAA Attestation of Excellence & Integrity (2023)** — Awarded by the Director General for outstanding operational performance & ethics.
- Technical Publications on DEV Community (2026)** — Author of architectural deep-dives: Zero-Hallucination GraphRAG & Dataset Automator MLOps.
- Google Cloud #AllThingsAgentic Hackathon** — Dataset Automator v4.0 (Google Antigravity, TabFM, BigQuery DataFrames, WIT).
- Google Developer Program · AICC Accra** — Active member · Accra AI Community Centre & Google for Startups Accelerator Network.

CONTACT & PROFILES

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📍 Douala / Ngaoundéré, CM

📄 github.com/gervais-afk

🔗 dev.to/gervais_marie

🌐 linkedin.com/in/marie-gervais-koa

AI & LLM STACK

Google Antigravity IDE ■ ■ ■ ■ ■

LangGraph & CrewAI ■ ■ ■ ■ ■

Google Gemma 4 (12B) ■ ■ ■ ■ ■

Gemini 2.5 / 1.5 Pro ■ ■ ■ ■ ■

Google TabFM (Tabular) ■ ■ ■ ■ ■

FastMCP & Genkit ■ ■ ■ ■ ■

Neo4j GraphRAG — Agent K1

DATA & GRAPHS

Neo4j / Cypher Graph ■ ■ ■ ■ ■

PostgreSQL / pgvector ■ ■ ■ ■ ■

Apache AGE & DuckDB ■ ■ ■ ■ ■

BigQuery DataFrames ■ ■ ■ ■ ■

Pandas / NumPy ETL ■ ■ ■ ■ ■

DEV & MLOPS

Python 3.11+ / MLOps ■ ■ ■ ■ ■

FastAPI / Next.js 14 ■ ■ ■ ■ ■

MLflow & Data Drift ■ ■ ■ ■ ■

SHAP Sentinel Audit ■ ■ ■ ■ ■

IfcOpenShell & Shapely ■ ■ ■ ■ ■

Docker & Pytest ■ ■ ■ ■ ■

ETHICS, SECURITY & AUDIT

◆ CCAA Excellence Attestation '23

◆ AVSEC Officer (ICAO Annex 17)

◆ AI Ethics & Anti-Hallucination

◆ OKF v0.2 SHA-256 No-LLM

◆ EU AI Act Compliance (RSASSA)

LANGUAGES

French — Native / Fluent

English — Tech & Written / Spoken Interm.